0. Definitions *coordinates, slope a'_{hd}, tracking error, gain measurement a_{xm}*

$$h[k] = h_d[k] - \delta h[k], \qquad \Delta h_d[k] = h_d[k+1] - h_d[k], \qquad \delta h_1 := \delta h[k-d], \quad d = 2$$

$$a'_{hd}[k] := \left.\frac{da_h}{dh}\right|_{h_d[k]}, \qquad a_r[k] := a_h[k] - a_h\big(h_d[k]\big)$$

$$\delta h[k+1] = \lambda_c\, \delta h[k] - F_{dh}[k]\, e_{a_h}[k] - \varepsilon_h[k]$$

$$a_{xm}[k] = a_h[k-d] + n_a[k], \qquad \mathrm{Var}(n_a) = R_2^{\mathrm{int}}, \qquad g_{\mathrm{hp}} := \frac{1-a_{var}}{a_{var}}, \qquad \beta = 1 - \frac{T_s}{\tau}$$

1. True *residual $= -a'_{hd}\,\delta h$; the measurement carries it delayed; the β-EWMA is the slope law*

$$a_h\big(h[k]\big) \approx a_h\big(h_d[k]\big) - a'_{hd}[k]\,\delta h[k]$$

$$\boxed{a_r[k] = -\,a'_{hd}[k]\,\delta h[k], \qquad a'_{hd}[k+1] \approx a'_{hd}[k]}$$

$$a'_{hdm}[k] := +\sqrt{\mathrm{Var}\big(a_r[k]\big)\,/\,\mathrm{Var}\big(\delta h[k]\big)} = a'_{hd}[k] \qquad (a'_{hd} > 0)$$

$$A[k] := a_h\big(h_d[k-d]\big), \qquad a_{xm}[k] = A[k] - a'_{hd}\,\delta h_1[k] + n_a[k]$$

$$\boxed{a'_{hd}[k+1] = \beta\, a'_{hd}[k] + (1-\beta)\, a'_{hdm}[k], \qquad a'_{hdm} \equiv a'_{hd}}$$

2. $\hat{a}'_{hdm}$ *measurement numerator: anchor-lag pre-compensation, floor subtraction, delay-aligned denominator*

$$\overline{a}_{xm}[k+1] = (1-a_{var})\,\overline{a}_{xm}[k] + a_{var}\,a_{xm}[k+1], \quad a_{xm,r}[k] = a_{xm}[k] - \overline{a}_{xm}[k]$$

$$\overline{\delta\hat{h}_1}[k+1] = (1-a_{var})\,\overline{\delta\hat{h}_1}[k] + a_{var}\,\delta\hat{h}_1[k+1], \qquad \delta\hat{h}_{1,r}[k] = \delta\hat{h}_1[k] - \overline{\delta\hat{h}_1}[k]$$

$$a_{xm,r}[k] = \mathrm{HP}[A][k] \;-\; a'_{hd}\,\delta h_1[k] \;+\; n_a[k]$$

$$\boxed{\mathrm{HP}[A][k] \approx g_{\mathrm{hp}}\,\frac{dA}{dk} = g_{\mathrm{hp}}\,a'_{hd}[k]\,\Delta h_d[k]}$$

$$\boxed{a^{\mathrm{corr}}_{xm,r}[k] = a_{xm,r}[k] - g_{\mathrm{hp}}\,\hat{a}'_{hd}[k]\,\Delta h_d[k] = g_{\mathrm{hp}}\,e_{a'_{hd}}[k]\,\Delta h_d[k] \;-\; a'_{hd}\,\delta h_1[k] \;+\; n_a[k]}$$

$$\hat{\sigma}^2_{a_{xm,r}}[k{+}1] = (1{-}a_{cov})\,\hat{\sigma}^2_{a_{xm,r}}[k] {+} a_{cov}\left(a^{\mathrm{corr}}_{xm,r}[k{+}1]\right)^2, \qquad \hat{\sigma}^2_{\delta\hat{h}_1}[k{+}1] = (1{-}a_{cov})\,\hat{\sigma}^2_{\delta\hat{h}_1}[k] {+} a_{cov}\,\delta\hat{h}^2_{1,r}[k{+}1]$$

$$\boxed{\hat{a}'_{hdm}[k] := +\sqrt{\left(\hat{\sigma}^2_{a_{xm,r}}[k] - R_2^{\mathrm{int}}[k]\right)^+ \,/\, \hat{\sigma}^2_{\delta\hat{h}_1}[k]}}$$

$$e_{a'_{hd}} = 0 \;\Rightarrow\; \hat{\sigma}^2_{a_{xm,r}} - R_2^{\mathrm{int}} = {a'_{hd}}^2\,\sigma^2_{\delta h} \;\Rightarrow\; \hat{a}'_{hdm} = a'_{hd}/\sqrt{\zeta} \approx a'_{hd}, \qquad \zeta := \mathrm{Var}(\delta\hat{h}_1)/\mathrm{Var}(\delta h) \approx 1$$

3. Esti (co-update) *a'_{hd} is driven by the β-EWMA of $\hat a'_{hdm}$* plus *the KF innovation correction*

$$x = [\,\delta h_1,\ \delta h_2,\ \delta h_3,\ a_h,\ a'_{hd}\,]^{\mathsf T}, \qquad K = P^- H^{\mathsf T} S^{-1}, \quad S = HP^- H^{\mathsf T} + R$$

$$\hat a_h[k+1] = \hat a_h[k] + \hat a'_{hd}[k]\big(\Delta h_d[k] + (1-\lambda_c)\,\delta\hat h_3[k]\big)$$

$$\boxed{\hat a'_{hd}[k+1] = \underbrace{\beta\,\hat a'_{hd}[k] + (1-\beta)\,\hat a'_{hdm}[k]}_{\text{var-ratio }\beta\text{-EWMA}} \ + \ \underbrace{L_{51}[k]\,e_{y_1}[k] + L_{52}[k]\,e_{y_2}[k]}_{\text{KF } L\cdot e}}$$

$$L_{5j} = [P^- H^{\mathsf T} S^{-1}]_{5,j} \ \propto\ P^-_{5,4} \ \ (\text{cross-cov, since } H(\cdot,5)=0), \qquad P^-_{5,4} \text{ built only by } F_e(4,5) = \Delta h_d$$

4. Error *$\kappa = 1$: the readout carries a genuine error, not the $\hat a_h$ echo ($\kappa = 0$)*

$$e_{a_h} = a_h - \hat a_h, \quad e_{a'_{hd}} = a'_{hd} - \hat a'_{hd}, \quad e_h = \delta h - \delta\hat h$$

$$e_{a_h}[k+1] = \big(1 + \hat a'_{hd} F_{dh}\big)\,e_{a_h} + \Delta h_d\, e_{a'_{hd}} + (1-\lambda_c)\,\hat a'_{hd}\,e_h + \hat a'_{hd}\,\varepsilon_h$$

$$\boxed{e_{a'_{hd}}[k+1] = \beta\, e_{a'_{hd}}[k] \ - \ (1-\beta)\big(\hat a'_{hdm}[k] - a'_{hdm}[k]\big) \ - \ (\text{KF correction}), \qquad \hat a'_{hdm} - a'_{hdm} = (\kappa{=}1)\ \text{error} + n_{var}}$$

5. $\mathbf{F}_e$ and H *$F_e(4,5) = \Delta h_d$ is the only path giving a'_{hd} a Kalman gain*

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh}[k] & 0 \\ 0 & 0 & (1-\lambda_c)\hat a'_{hd}[k] & 1 + \hat a'_{hd}[k] F_{dh}[k] & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & \beta \end{bmatrix}, \qquad H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

6. Q and R *variance-ratio readout noise is* quadratic *in the floor*

$$Q = \operatorname{diag}\big(0,\ 0,\ Q_{33},\ Q_{44},\ Q_{55}\big), \qquad R = \operatorname{diag}\big(\sigma_n^2,\ R_2^{\text{int}}\big), \qquad Q_{55} = (1-\beta)^2 R_{var}$$

$$Q_{33} = 4k_B T\,\hat a_h\big(1{+}(1{-}\lambda_c)^2 d\big){+}(1{-}\lambda_c)^2\sigma_n^2, \qquad Q_{44} = \big(\hat a_h K_h/R\big)^2\,\sigma_{\delta h}^2, \qquad R_2^{\text{int}} = a_{cov}\,IF_{var}\,\big(\hat a_h{+}\xi\big)^2$$

$$\boxed{R_{var} \ \approx\ \frac{a_{cov}\,IF}{4}\ \frac{\big(R_2^{\text{int}}\big)^2}{{a'_{hd}}^2\,\sigma_{\delta h}^4} \qquad (\text{noise } \textit{squared})}$$

colored-a_{xm} : a_{xm} is an IIR-variance output (slow) $\Rightarrow$ its high-pass keeps only a fraction of R_2^{int} $\Rightarrow$ full-R_2^{int} su

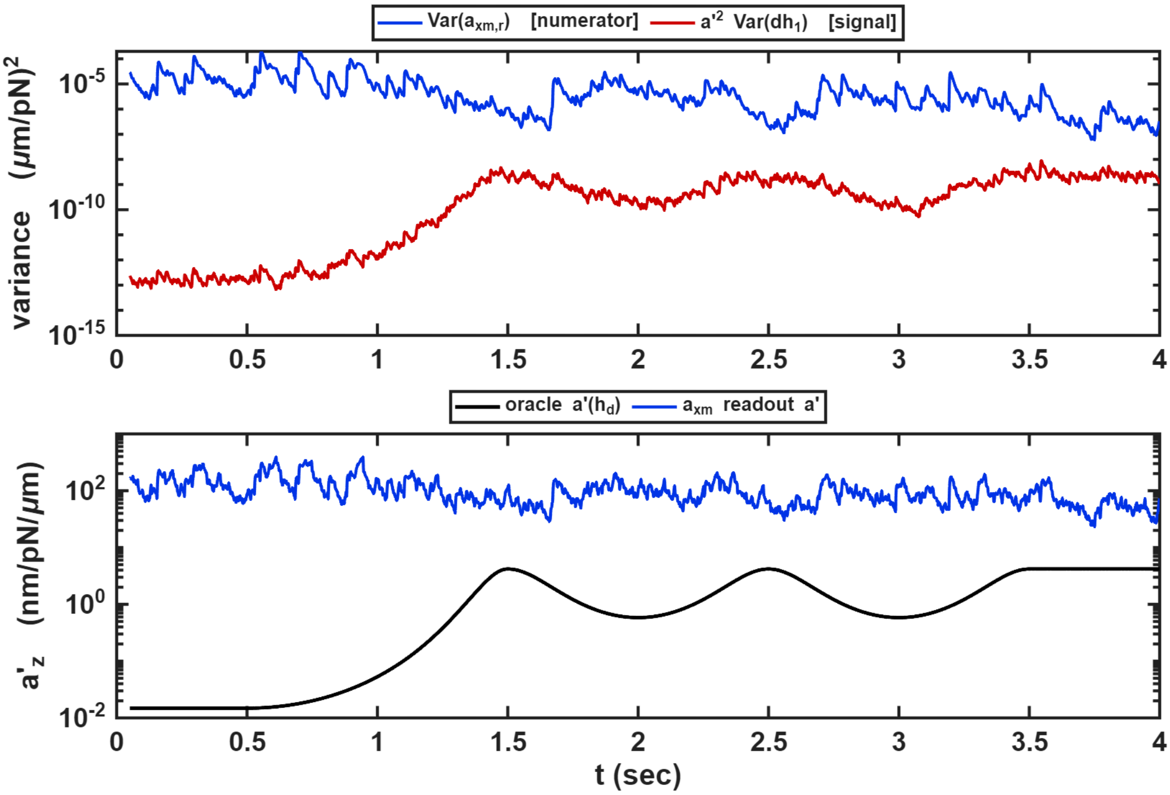
7. Regression (noise-linear limit) *optimizing the compensation over $\hat{a}'_{hd}$ drops the floor from quadratic to linear*

$$\hat{a}'_{hd}{}^{\text{reg}}[k] = \frac{\mathrm{Cov}\big(a_{xm,r},\, \Delta h_d\big)}{g_{\text{hp}}\,\mathrm{Var}\big(\Delta h_d\big)}, \qquad \mathrm{Cov}\big(a_{xm,r}, \Delta h_d\big) = g_{\text{hp}}\, a'_{hd}\,\mathrm{Var}(\Delta h_d)\ \underbrace{+\,0}_{-a'_{hd}\delta h_1}\ \underbrace{+\,0}_{n_a}$$

$$\boxed{\hat{a}'_{hd}{}^{\text{reg}} = a'_{hd}, \qquad R_{var}^{\text{reg}} \;\approx\; \frac{R_2^{\text{int}}}{N_{\text{eff}}\; g_{\text{hp}}^2\,\mathrm{Var}(\Delta h_d)} \qquad (\text{noise } \textit{linear})}$$

$R_{var}^{\text{reg}} \propto R_2^{\text{int}}$ vs $R_{var} \propto (R_2^{\text{int}})^2$; but drive $g_{\text{hp}}a'_{hd}\Delta h_d \ll n_a$, and $\Delta h_d \to 0$ twice/cycle.

8. Simulation *open-loop readout: the numerator cannot read a'_{hd}*



Open-loop a_{xm} numerator (taylor 4-state, seed 1). Top: the numerator variance $\mathrm{Var}(a_{xm,r})$ (blue) sits $\sim 5\times10^3$ above the thermal signal $a'_{hd}{}^2\mathrm{Var}(\delta h)$ (red) it should equal — the a_{xm} noise floor. Bottom: the readout $\hat{a}'_{hdm} = \sqrt{\cdot}$ is therefore $\sim$70$\times$ oracle $a'_{hd}(h_d)$ (black).

9. Verdict *$\kappa = 1$, truth is a fixed point — but the channel is below its own resolution*

$$\boxed{\hat{a}'_{hdm}/a'_{hd} \approx 70, \qquad \big(\hat{a}'_{hdm}/a'_{hd}\big)^2 = \frac{\mathrm{Var}(a_{xm,r})}{a'_{hd}{}^2\,\sigma_{\delta h}^2} \approx 5\times10^3 \quad (\text{osc, near wall})}$$

Structurally sound ($\kappa = 1$, fixed point at a'_{hd}, no $c(\bar{h})$) but information-limited: the readout is buried in the a_{xm}

Past the floor $\Rightarrow$ more information: trajectory dither (Δh_d non-zero-crossing) or an independent c-free channel.